BoardNet
Complete Configuration Commands

Board Examination Communication Network | Cisco Packet Tracer

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1. Addressing Table

Segment	Network	Mask	Assigned addresses
DHK LAN	15.13.0.0/23	255.255.254.0	Gateway 15.13.0.1
CTG LAN	15.13.2.0/24	255.255.255.0	Gateway 15.13.2.1
RAJ LAN	15.13.3.0/24	255.255.255.0	Gateway 15.13.3.1
SYL LAN	15.13.4.0/24	255.255.255.0	Gateway 15.13.4.1
KHU LAN	15.13.5.0/25	255.255.255.128	Gateway 15.13.5.1
BAR LAN	15.13.5.128/25	255.255.255.128	Gateway 15.13.5.129
DHK-SYL-RAJ Central	15.13.6.0/29	255.255.255.248	DHK .1, SYL .2, RAJ .3
DHK-KHU	15.13.6.8/30	255.255.255.252	DHK .9, KHU .10
KHU-BAR	15.13.6.12/30	255.255.255.252	KHU .13, BAR .14
BAR-DHK	15.13.6.16/30	255.255.255.252	BAR .17, DHK .18
DHK-CTG	15.13.6.20/30	255.255.255.252	DHK .21, CTG .22

2. Router Interface Configuration

2.1 DHK-R

```
enable
configure terminal
hostname DHK-R

interface gigabitEthernet0/0
ip address 15.13.0.1 255.255.254.0
no shutdown
exit

interface gigabitEthernet0/1
ip address 15.13.6.1 255.255.255.248
no shutdown
exit

interface gigabitEthernet0/2
ip address 15.13.6.21 255.255.255.252
no shutdown
exit

interface serial0/2/0
ip address 15.13.6.9 255.255.255.252
no shutdown
exit

interface serial0/2/1
ip address 15.13.6.18 255.255.255.252
no shutdown
exit

end
show ip interface brief
```

2.2 CTG-R

```
enable
configure terminal
hostname CTG-R

interface gigabitEthernet0/0
ip address 15.13.2.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
ip address 15.13.6.22 255.255.255.252
no shutdown
exit

ip route 0.0.0.0 0.0.0.0 15.13.6.21

end
show ip interface brief
```

2.3 SYL-R

```
enable
configure terminal
hostname SYL-R

interface gigabitEthernet0/0
ip address 15.13.4.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
ip address 15.13.6.2 255.255.255.248
no shutdown
exit

end
show ip interface brief
```

2.4 RAJ-R

```
enable
configure terminal
hostname RAJ-R

interface gigabitEthernet0/0
ip address 15.13.3.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
ip address 15.13.6.3 255.255.255.248
no shutdown
exit

end
show ip interface brief
```

2.5 KHU-R

```
enable
configure terminal
hostname KHU-R

interface gigabitEthernet0/0
ip address 15.13.5.1 255.255.255.128
```

```
no shutdown
exit

interface serial0/3/0
ip address 15.13.6.10 255.255.255.252
no shutdown
exit

interface serial0/3/1
ip address 15.13.6.13 255.255.255.252
no shutdown
exit

end
show ip interface brief
```

2.6 BAR-R

```
enable
configure terminal
hostname BAR-R

interface gigabitEthernet0/0
ip address 15.13.5.129 255.255.255.128
no shutdown
exit

interface serial0/3/0
no ip address
ip address 15.13.6.17 255.255.255.252
no shutdown
exit

interface serial0/3/1
no ip address
ip address 15.13.6.14 255.255.255.252
no shutdown
exit

end
show ip interface brief
```

2.7 Optional DCE clock-rate check

Only if a serial link is up/down or not coming up.

```
show controllers serial 0/2/0
configure terminal
interface serial0/2/0
clock rate 64000
no shutdown
exit
end
```

3. Direct Connectivity Verification

3.1 DHK direct neighbors

```
ping 15.13.6.2
ping 15.13.6.3
ping 15.13.6.22
ping 15.13.6.10
ping 15.13.6.17
```

3.2 KHU / BAR triangle checks

```
KHU-R# ping 15.13.6.9
KHU-R# ping 15.13.6.14

BAR-R# ping 15.13.6.13
BAR-R# ping 15.13.6.18

DHK-R# ping 15.13.6.10
DHK-R# ping 15.13.6.17
```

3.3 CDP verification used to confirm BAR link mapping

```
show cdp neighbors
```

4. Routing Configuration

Routing policy implemented: CTG uses a default route; DHK/SYL/RAJ use RIP v2; KHU and BAR use static routes with floating backups for redundancy.

4.1 DHK-R static routes + RIP v2

```
enable
configure terminal

ip route 15.13.2.0 255.255.255.0 15.13.6.22

ip route 15.13.5.0 255.255.255.128 15.13.6.10
ip route 15.13.5.0 255.255.255.128 15.13.6.17 5

ip route 15.13.5.128 255.255.255.128 15.13.6.17
ip route 15.13.5.128 255.255.255.128 15.13.6.10 5

ip route 15.13.6.12 255.255.255.252 15.13.6.10
ip route 15.13.6.12 255.255.255.252 15.13.6.17 5

router rip
version 2
no auto-summary
network 15.0.0.0
passive-interface gigabitEthernet0/0
passive-interface gigabitEthernet0/2
passive-interface serial0/2/0
passive-interface serial0/2/1
redistribute static metric 1
exit

end
show ip route
```

4.2 SYL-R RIP v2

```
enable
configure terminal
router rip
version 2
no auto-summary
network 15.0.0.0
passive-interface gigabitEthernet0/0
exit
end
show ip route
```

4.3 RAJ-R RIP v2

```
enable
configure terminal
router rip
version 2
no auto-summary
network 15.0.0.0
passive-interface gigabitEthernet0/0
exit
end
show ip route
```

4.4 KHU-R static + floating static routes

```
enable
configure terminal

ip route 15.13.0.0 255.255.254.0 15.13.6.9
ip route 15.13.0.0 255.255.254.0 15.13.6.14 5

ip route 15.13.2.0 255.255.255.0 15.13.6.9
ip route 15.13.2.0 255.255.255.0 15.13.6.14 5

ip route 15.13.3.0 255.255.255.0 15.13.6.9
ip route 15.13.3.0 255.255.255.0 15.13.6.14 5

ip route 15.13.4.0 255.255.255.0 15.13.6.9
ip route 15.13.4.0 255.255.255.0 15.13.6.14 5

ip route 15.13.5.128 255.255.255.128 15.13.6.14
ip route 15.13.5.128 255.255.255.128 15.13.6.9 5

ip route 15.13.6.0 255.255.255.248 15.13.6.9
ip route 15.13.6.0 255.255.255.248 15.13.6.14 5

ip route 15.13.6.20 255.255.255.252 15.13.6.9
ip route 15.13.6.20 255.255.255.252 15.13.6.14 5

end
show ip route
```

4.5 BAR-R static + floating static routes

```
enable
configure terminal

ip route 15.13.0.0 255.255.254.0 15.13.6.18
ip route 15.13.0.0 255.255.254.0 15.13.6.13 5

ip route 15.13.2.0 255.255.255.0 15.13.6.18
ip route 15.13.2.0 255.255.255.0 15.13.6.13 5

ip route 15.13.3.0 255.255.255.0 15.13.6.18
ip route 15.13.3.0 255.255.255.0 15.13.6.13 5

ip route 15.13.4.0 255.255.255.0 15.13.6.18
ip route 15.13.4.0 255.255.255.0 15.13.6.13 5

ip route 15.13.5.0 255.255.255.128 15.13.6.13
ip route 15.13.5.0 255.255.255.128 15.13.6.18 5

ip route 15.13.6.0 255.255.255.248 15.13.6.18
ip route 15.13.6.0 255.255.255.248 15.13.6.13 5

ip route 15.13.6.20 255.255.255.252 15.13.6.18
ip route 15.13.6.20 255.255.255.252 15.13.6.13 5
```

```
end
show ip route
```

4.6 Routing verification pings

```
SYL-R# ping 15.13.5.1
RAJ-R# ping 15.13.5.129
KHU-R# ping 15.13.2.1
BAR-R# ping 15.13.4.1
CTG-R# ping 15.13.5.129
```

5. DHCP Configuration

5.1 DHK-R DHCP server - four pools

```
enable
configure terminal

ip dhcp excluded-address 15.13.0.1 15.13.0.20
ip dhcp excluded-address 15.13.2.1 15.13.2.20
ip dhcp excluded-address 15.13.3.1 15.13.3.20
ip dhcp excluded-address 15.13.4.1 15.13.4.20

ip dhcp pool DHK
network 15.13.0.0 255.255.254.0
default-router 15.13.0.1
dns-server 15.13.0.2
exit

ip dhcp pool CTG
network 15.13.2.0 255.255.255.0
default-router 15.13.2.1
dns-server 15.13.0.2
exit

ip dhcp pool RAJ
network 15.13.3.0 255.255.255.0
default-router 15.13.3.1
dns-server 15.13.0.2
exit

ip dhcp pool SYL
network 15.13.4.0 255.255.255.0
default-router 15.13.4.1
dns-server 15.13.0.2
exit

end
show ip dhcp pool
show ip dhcp binding
```

5.2 CTG-R DHCP relay

```
enable
configure terminal
interface gigabitEthernet0/0
ip helper-address 15.13.6.21
exit
end
show ip interface gigabitEthernet0/0
```

5.3 SYL-R DHCP relay

```
enable
configure terminal
interface gigabitEthernet0/0
ip helper-address 15.13.6.1
exit
end
show ip interface gigabitEthernet0/0
```

5.4 RAJ-R DHCP relay

```
enable
configure terminal
interface gigabitEthernet0/0
ip helper-address 15.13.6.1
exit
end
show ip interface gigabitEthernet0/0
```

5.5 KHU dedicated DHCP Server (GUI)

Click path: KHU-DHCP -> Desktop -> IP Configuration -> Static

Field	Value
IP Address	15.13.5.3
Subnet Mask	255.255.255.128
Default Gateway	15.13.5.1
DNS Server	15.13.0.2

Then: Services -> DHCP -> On -> Add the pool below.

DHCP Field	Value
Pool Name	KHU
Default Gateway	15.13.5.1
DNS Server	15.13.0.2
Start IP Address	15.13.5.20
Subnet Mask	255.255.255.128
Maximum Users	100

5.6 BAR dedicated DHCP Server (GUI)

Click path: BAR-DHCP -> Desktop -> IP Configuration -> Static

Field	Value
IP Address	15.13.5.131
Subnet Mask	255.255.255.128
Default Gateway	15.13.5.129
DNS Server	15.13.0.2

Then: Services -> DHCP -> On -> Add the pool below.

DHCP Field	Value
Pool Name	BAR
Default Gateway	15.13.5.129
DNS Server	15.13.0.2
Start IP Address	15.13.5.150
Subnet Mask	255.255.255.128
Maximum Users	100

6. Static Server and Printer Addressing

All routers, servers, and printers are static. Only PCs use DHCP.

Device	IP	Mask	Gateway	DNS
DHK-DNS	15.13.0.2	255.255.254.0	15.13.0.1	15.13.0.2
DHK-WEB	15.13.0.3	255.255.254.0	15.13.0.1	15.13.0.2
DHK-MAIL	15.13.0.4	255.255.254.0	15.13.0.1	15.13.0.2
DHK-Printer	15.13.0.5	255.255.254.0	15.13.0.1	-
CTG-MAIL	15.13.2.2	255.255.255.0	15.13.2.1	15.13.0.2
CTG-WEB	15.13.2.3	255.255.255.0	15.13.2.1	15.13.0.2
CTG-Printer	15.13.2.4	255.255.255.0	15.13.2.1	-
RAJ-MAIL	15.13.3.2	255.255.255.0	15.13.3.1	15.13.0.2
RAJ-Printer	15.13.3.3	255.255.255.0	15.13.3.1	-
SYL-MAIL	15.13.4.2	255.255.255.0	15.13.4.1	15.13.0.2
SYL-Printer	15.13.4.3	255.255.255.0	15.13.4.1	-
KHU-MAIL	15.13.5.2	255.255.255.128	15.13.5.1	15.13.0.2
KHU-DHCP	15.13.5.3	255.255.255.128	15.13.5.1	15.13.0.2
KHU-Printer	15.13.5.4	255.255.255.128	15.13.5.1	-
BAR-MAIL	15.13.5.130	255.255.255.128	15.13.5.129	15.13.0.2
BAR-DHCP	15.13.5.131	255.255.255.128	15.13.5.129	15.13.0.2
BAR-Printer	15.13.5.132	255.255.255.128	15.13.5.129	-

7. DNS Configuration

Click path: DHK-DNS -> Services -> DNS -> On. Add these A records one at a time.

DNS Name	Address
www.dhk.edu.bd	15.13.0.3
www.ctg.edu.bd	15.13.2.3
mail.dhk.edu.bd	15.13.0.4
mail.ctg.edu.bd	15.13.2.2
mail.raj.edu.bd	15.13.3.2
mail.syl.edu.bd	15.13.4.2
mail.khu.edu.bd	15.13.5.2
mail.bar.edu.bd	15.13.5.130

8. Web Server Configuration

8.1 DHK-WEB

Click path: DHK-WEB -> Services -> HTTP -> HTTP On.

```
Optional index.html content:
<h1>Dhaka Board Examination Portal</h1>
```

8.2 CTG-WEB

Click path: CTG-WEB -> Services -> HTTP -> HTTP On.

```
Optional index.html content:
<h1>Chattogram Board Examination Portal</h1>
```

8.3 Web test

On any PC: Desktop -> Web Browser, then open:

```
http://www.dhk.edu.bd
http://www.ctg.edu.bd
```

9. Email Server Configuration

On every mail server: Services -> EMAIL -> SMTP On, POP3 On. Configure the domain and users below.

Server	Domain	Users
DHK-MAIL	dhk.edu.bd	dhk1 / 123; dhk2 / 123
CTG-MAIL	ctg.edu.bd	ctg1 / 123; ctg2 / 123
RAJ-MAIL	raj.edu.bd	raj1 / 123; raj2 / 123
SYL-MAIL	syl.edu.bd	syl1 / 123; syl2 / 123
KHU-MAIL	khu.edu.bd	khu1 / 123; khu2 / 123
BAR-MAIL	bar.edu.bd	bar1 / 123; bar2 / 123

9.1 Example PC email account - DHK-PC1

Click path: DHK-PC1 -> Desktop -> Email

Field	Value
Your Name	DHK PC1
Email Address	dhk1@dhk.edu.bd
Incoming Mail Server	mail.dhk.edu.bd
Outgoing Mail Server	mail.dhk.edu.bd
Username	dhk1
Password	123

9.2 Example cross-board email test

From DHK-PC1 -> Desktop -> Email -> Compose:

```
To: ctg1@ctg.edu.bd
Subject: BoardNet Test
Message: Test email from DHK.
```

Then on CTG-PC1 -> Desktop -> Email -> Receive.

10. End-Device Setup and Service Testing

10.1 Put all PCs on DHCP

For every PC: Desktop -> IP Configuration -> DHCP. Then verify using:

```
ipconfig
```

10.2 Gateway tests

```
DHK-PC> ping 15.13.0.1
CTG-PC> ping 15.13.2.1
RAJ-PC> ping 15.13.3.1
SYL-PC> ping 15.13.4.1
KHU-PC> ping 15.13.5.1
BAR-PC> ping 15.13.5.129
```

10.3 DNS tests

```
ping www.dhk.edu.bd
ping www.ctg.edu.bd
ping mail.dhk.edu.bd
ping mail.khu.edu.bd
ping mail.bar.edu.bd
```

10.4 Representative cross-board tests

```
CTG-PC1> ping 15.13.0.2
CTG-PC1> ping 15.13.3.2
CTG-PC1> ping 15.13.4.2
```

```
CTG-PC1> ping 15.13.5.2
CTG-PC1> ping 15.13.5.130

KHU-PC1> ping 15.13.0.2
KHU-PC1> ping 15.13.2.3
KHU-PC1> ping 15.13.3.2
KHU-PC1> ping 15.13.4.2
KHU-PC1> ping 15.13.5.130

BAR-PC1> ping 15.13.0.3
BAR-PC1> ping 15.13.2.3
BAR-PC1> ping 15.13.3.2
BAR-PC1> ping 15.13.4.2
BAR-PC1> ping 15.13.5.2
```

11. Redundancy Testing

11.1 Test: DHK-KHU primary link down

```
DHK-R# configure terminal
DHK-R(config)# interface serial0/2/0
DHK-R(config-if)# shutdown
DHK-R(config-if)# end

KHU-R# show ip route 15.13.0.0

KHU-PC1> ping 15.13.0.2
```

Expected backup direction: KHU -> BAR -> DHK.

Restore the link:

```
DHK-R# configure terminal
DHK-R(config)# interface serial0/2/0
DHK-R(config-if)# no shutdown
DHK-R(config-if)# end
```

11.2 Test: DHK-BAR primary link down

```
DHK-R# configure terminal
DHK-R(config)# interface serial0/2/1
DHK-R(config-if)# shutdown
DHK-R(config-if)# end

BAR-R# show ip route 15.13.0.0

BAR-PC1> ping 15.13.0.2
```

Expected backup direction: BAR -> KHU -> DHK.

Restore the link:

```
DHK-R# configure terminal
DHK-R(config)# interface serial0/2/1
DHK-R(config-if)# no shutdown
DHK-R(config-if)# end
```

12. Save and Final Verification Commands

12.1 Save each router

```
enable
copy running-config startup-config
```

Alternative save command:

```
write memory
```

12.2 Useful final verification commands

```
show ip interface brief
show ip route
show ip dhcp pool
show ip dhcp binding
show running-config
show cdp neighbors
```